

Estimating PhonePe's Daily UPI Transactions

Using structured assumptions and market insights to estimate PhonePe's daily transaction volume in the absence of real-world data.

PRODUCT MANAGEMENT GUESSTIMATE

The Problem

This wasn't about finding the exact number. It was about answering a business question with limited information and showing a structured way of thinking.

The Question

How many UPI transactions does PhonePe process on an average day in India?

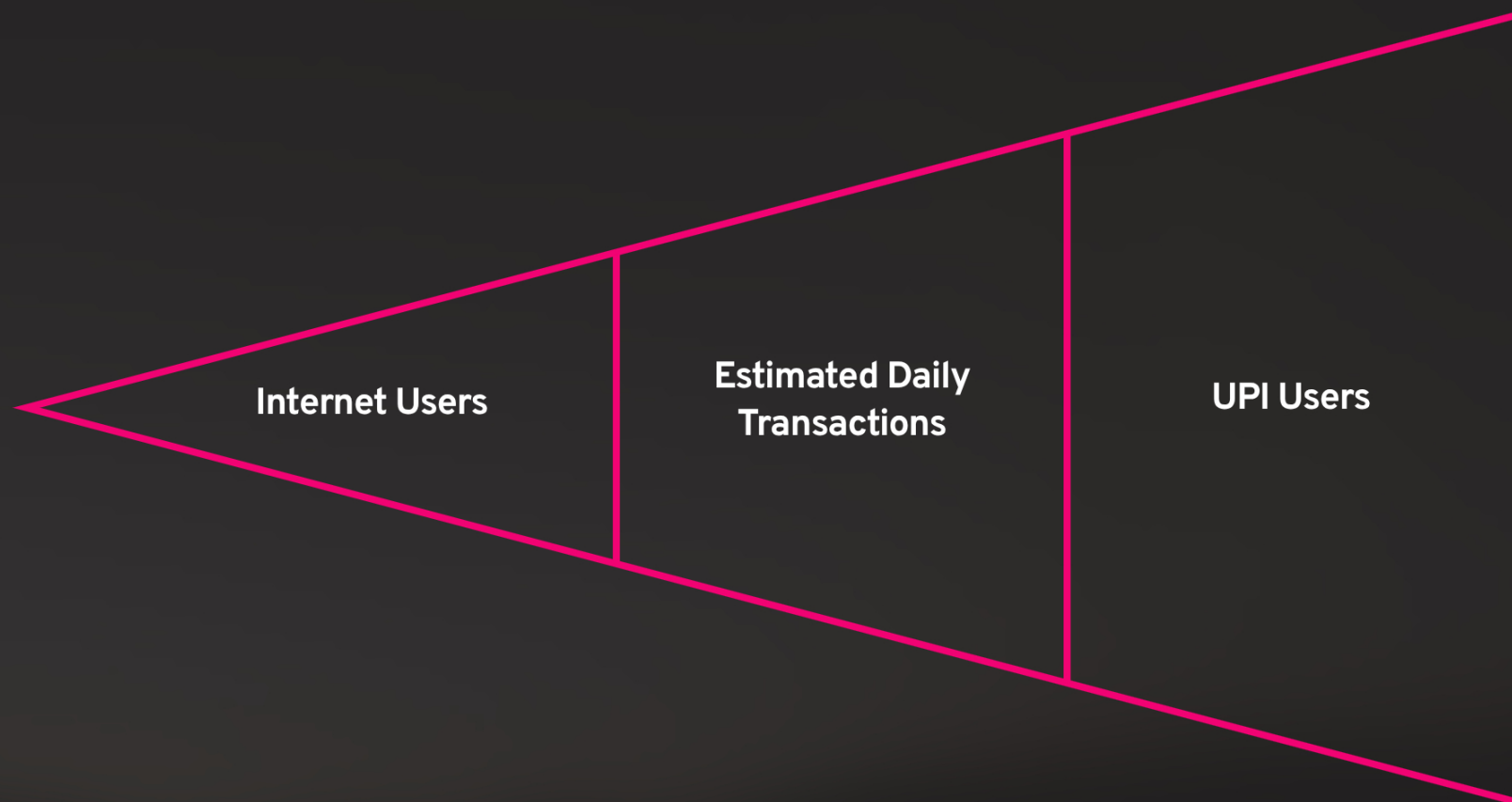
Scope

- India only
- UPI transactions only
- Include both P2P and merchant payments
- Assume an average business day (no festival spikes)

My Approach

Rather than making a random guess, I broke the problem into smaller assumptions that could be estimated logically – starting with India's population and progressively narrowing down to estimated daily transactions.

Formula: Daily Transactions = Active PhonePe Users × Average Transactions per Active User



This top-down funnel approach ensures every assumption is traceable and defensible – a hallmark of rigorous product thinking.

Key Assumptions

Since exact numbers aren't publicly available, I used realistic market assumptions grounded in publicly known benchmarks. These figures are rounded for estimation and demonstrate the approach rather than providing precise figures.

1.4B

India's Population

Total base for estimation

1B

Internet Users

Digitally reachable population

560M

UPI Users

Active across all platforms

50%

PhonePe Market Share

Estimated UPI market
dominance

280M

PhonePe Users

Estimated total user base

User Segmentation

Not every user pays the same way. Instead of using one average for everyone, I grouped users based on how frequently they use UPI. Segmenting users made the estimate far more realistic than assuming identical behavior across the entire base.

Daily Users

~3 transactions/day

Power users – street vendors, frequent shoppers, urban commuters relying on UPI for every payment.

Weekly Users

~2–3 transactions/week

Regular users making grocery runs, bill payments, or peer transfers a few times per week.

Monthly Users

~3 transactions/month

Occasional users who pay digitally only for specific needs like utility bills or rent.

Estimation

Applying the segmentation model to the estimated user base yields a bottom-up calculation for total daily transaction volume across all three cohorts.

Daily Active Users

~300 Million transactions

Weekly Active Users

~41 Million transactions

Monthly Active Users

~7 Million transactions

Estimated Daily Transaction Volume

~350M

UPI Transactions per day on PhonePe

Validating the Estimate

An estimate is only useful if it passes a reality check. I compared my result with publicly available NPCI transaction trends to test plausibility.

NPCI Benchmark

India processes roughly **600–630 million UPI transactions every day** across all platforms. With PhonePe holding ~50% market share, the expected range is **280–300 million** daily transactions.

My Estimate

My estimate of **~350 million** is slightly higher but still falls within a reasonable range given the assumptions used. Small variance is expected in any structured estimation exercise.

Key Learnings

This exercise reinforced three important Product Management skills that apply far beyond this single guesstimate – shaping how PMs approach ambiguous, data-scarce decisions every day.



Structured Decomposition

Breaking ambiguous problems into smaller, measurable assumptions makes any complex question tractable and defensible.



Informed Estimation

Making confident estimates with limited data is a core PM skill – grounding guesses in logic separates great PMs from average ones.



Reality Validation

Validating assumptions against real-world benchmarks before drawing conclusions ensures the estimate remains credible and trustworthy.